

Who buys insurance from our campaigns, and how to target the next one

Analysis of 214,493 PA and Life insurance offers: response drivers, product fit, and a propensity model for smarter targeting.

1.22%

acceptance rate

3×

better response from active accounts

49%

of buyers in the top 20% by model score

Response is low, but highly predictable, so we can contact fewer people and sell more

The four numbers and three messages to remember

Acceptance rate

1.22%

2,621 of 214,493 bought

■ PA / ■ Life

62 / 38

share of sales (0.76% / 0.46%)

Best group

2.1%

new accounts (1–12 months on book)

Model

49%

of buyers reached by calling the top 20%

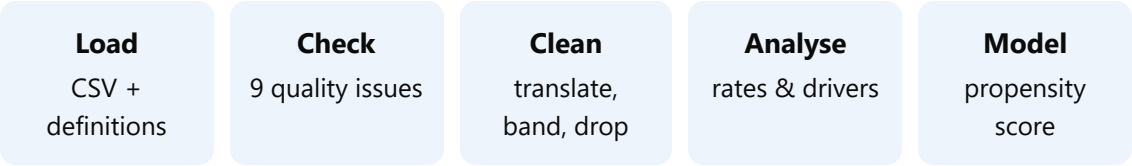
- 1 **Who buys:** young, active customers. Customers aged ≤ 25 , accounts with cash inflow and customers 1–36 months on book convert at 1.5–2.1%.
- 2 **What they buy:** PA for the mass market, **Life for the affluent** (Upper Mass, income $\geq 30k$, credit-card holders).
- 3 **How to target:** score customers before each campaign. Tiers A + B (50% of the list) hold **82% of the buyers**.

One campaign file, 26 fields, three outcomes, analysed with a reproducible pipeline

Every number in this deck comes from `python main.py`

Rows	215,993 offers (1 row = 1 customer offered)
Analysed	214,493 after removing 1,500 empty profiles
Fields	Demographics, segment, product holding, balances, cash flow, tenure
Outcome (label)	0 = Reject · 1 = PA · 2 = Life
Period	12 campaign months (Jan–Dec, no year)

Pipeline



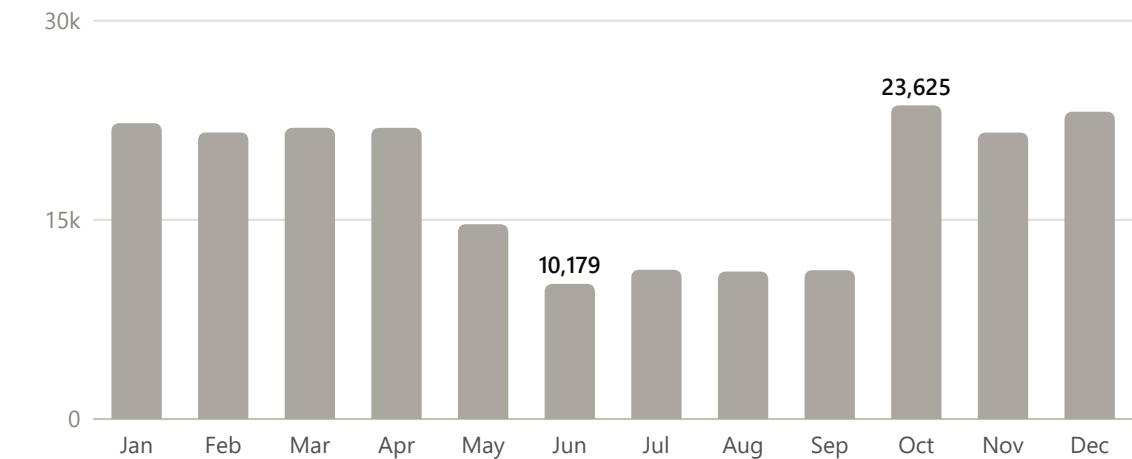
Data issues handled

- 15-day cash-flow fields are exact copies of the 30-day ones, so they were dropped
- Current-account balance is always 0, so it was dropped
- Marital status in Thai was translated and merged
- Only 1.2% of rows are positive, so the model is judged with AUC and gains, not accuracy

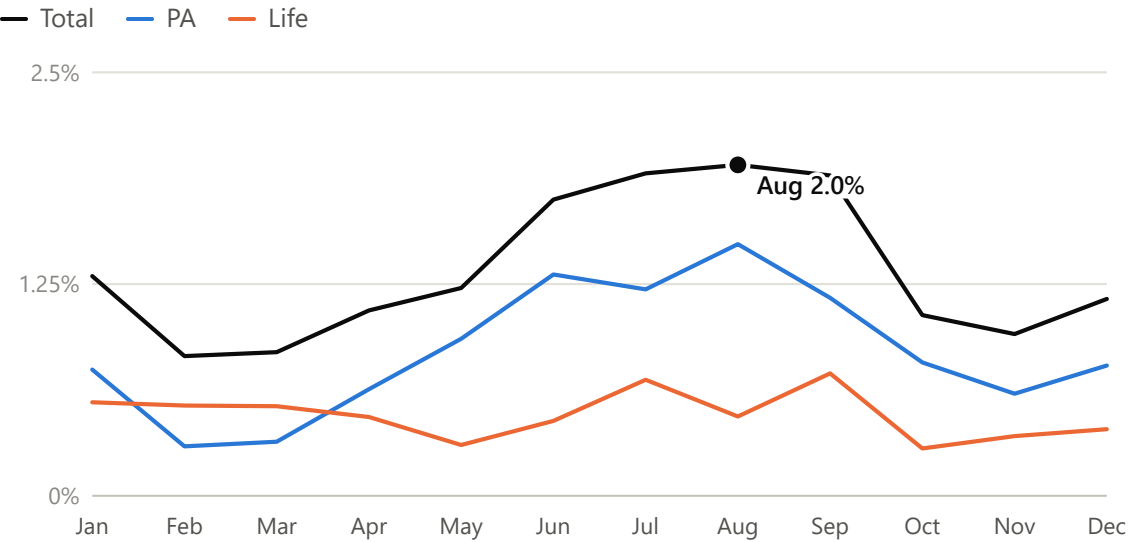
Smaller campaigns converted almost twice as well

Jun–Sep contacted ~11k customers a month and reached ~1.9%; the 22k-a-month campaigns reached 0.8–1.3%

Customers contacted per month



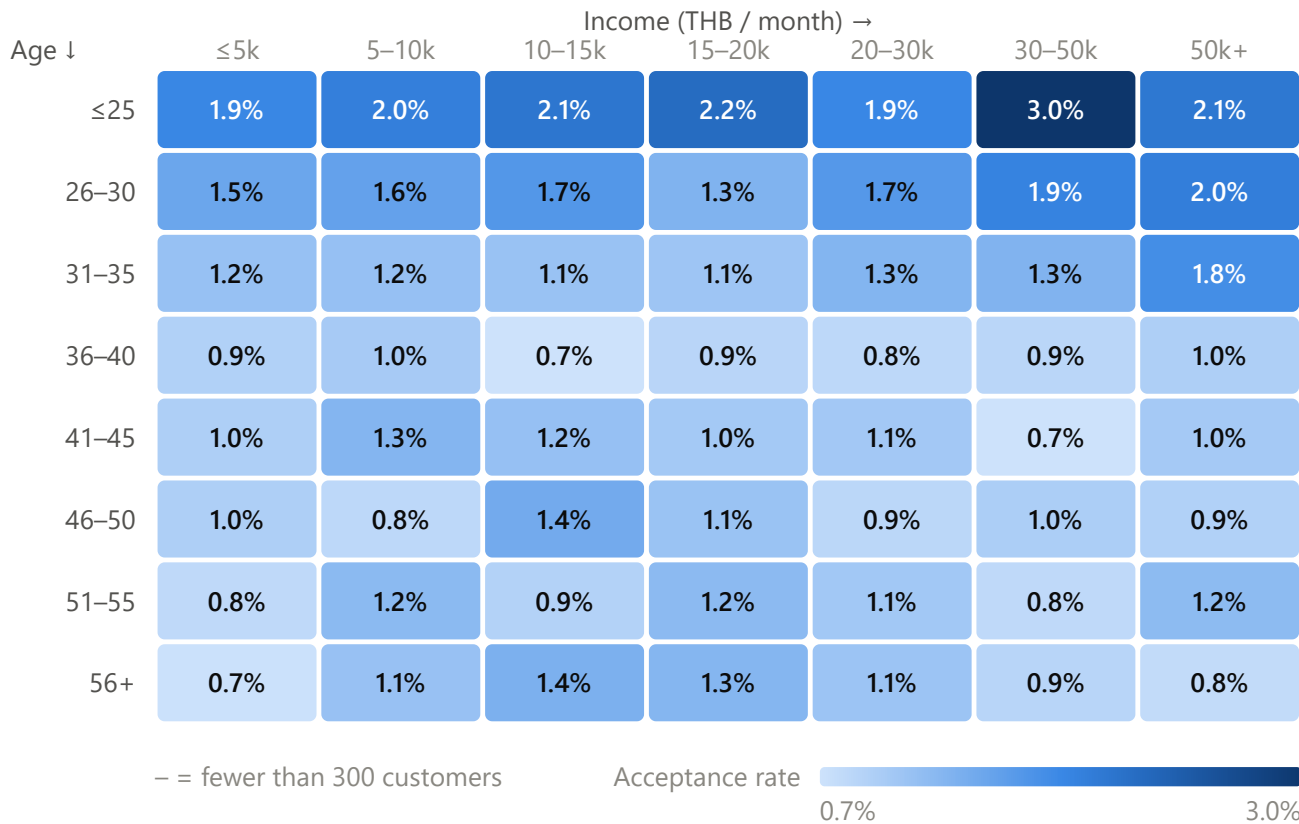
Acceptance rate per month



So what: more contacts did not bring proportionally more sales. The swing is almost all PA; Life stays at 0.3–0.7%. Contact fewer, better-chosen customers.

Age matters more than income: customers aged 25 and under convert at ~2% at every income level

Acceptance rate for each age × income combination



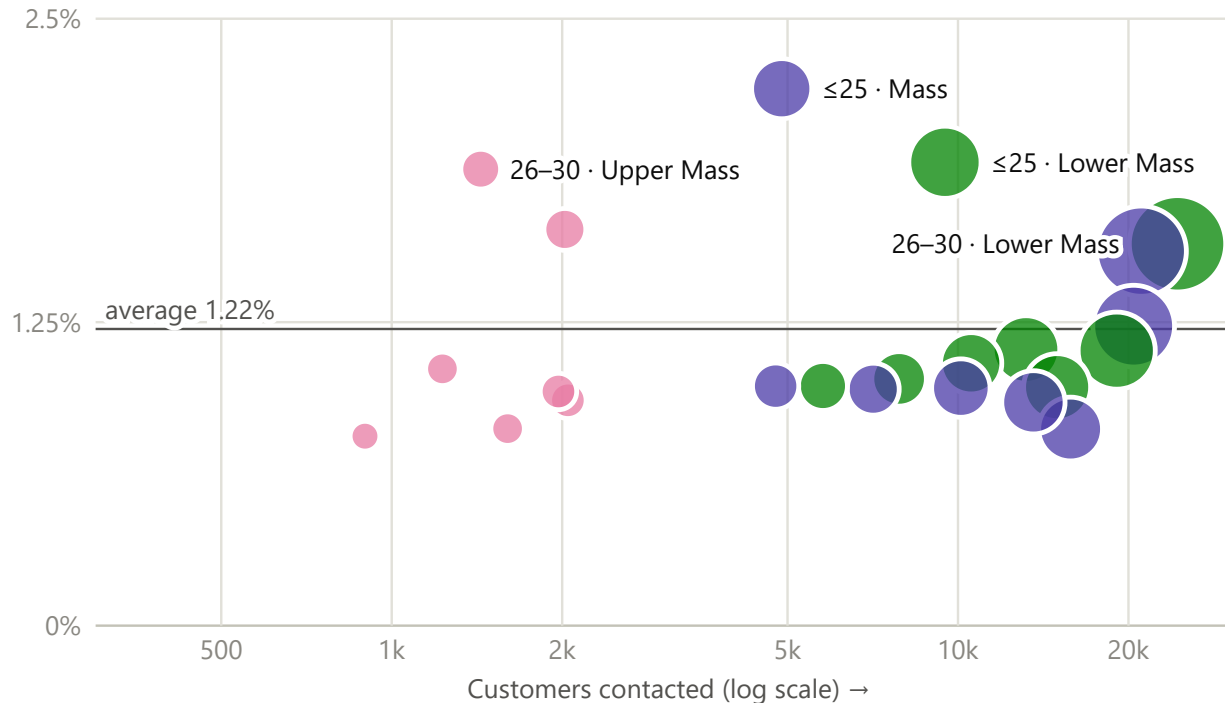
- 1 The top row (≤25) is dark across every income band: **1.9–3.0%**.
- 2 Above 35, almost every cell is **0.7–1.4%**, whatever the income.
- 3 The single best cell is age ≤25 with income 30–50k at **3.0%** (726 customers).

So what: lead the targeting with age. Income alone does not find buyers.

The biggest wins sit with young Lower Mass and Mass customers

Each bubble is an age band within a segment: x = customers contacted, y = acceptance, size = buyers

● Lower Mass ● Mass ● Upper Mass



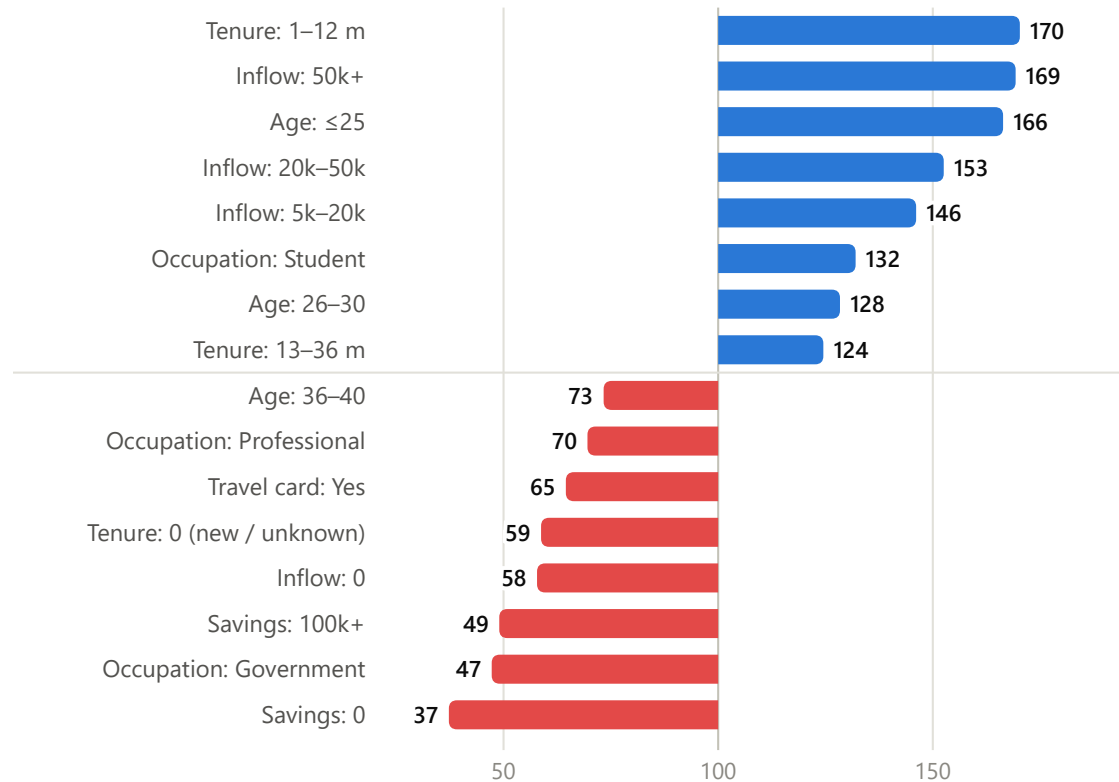
- 1 **Top-right = priority:** large groups that also convert above average. 26–30 Lower Mass alone gives 384 buyers at 1.57%.
- 2 **≤25 Mass** has the highest rate (2.2%), and ≤25 Lower Mass adds 181 buyers at 1.9%.
- 3 Upper Mass groups are small (left side) but young Upper Mass converts well (1.6–1.9%).

So what: spend contact effort above the average line, starting with the biggest bubbles.

Activity, tenure and age separate buyers from rejecters

Response index = group acceptance ÷ average × 100 · the 8 best and 8 worst groups (≥1,000 customers)

■ Above average ■ Below average



- 1 Best:** 1–12 months on book (170), inflow 50k+ (169), age ≤25 (166): **about 1.7× the average.**
- 2 Worst:** zero savings (37), Government (47), savings 100k+ (49): **about half the average or less.**
- 3** Most of the top 8 are about **account activity** (inflow, tenure), not demographics.

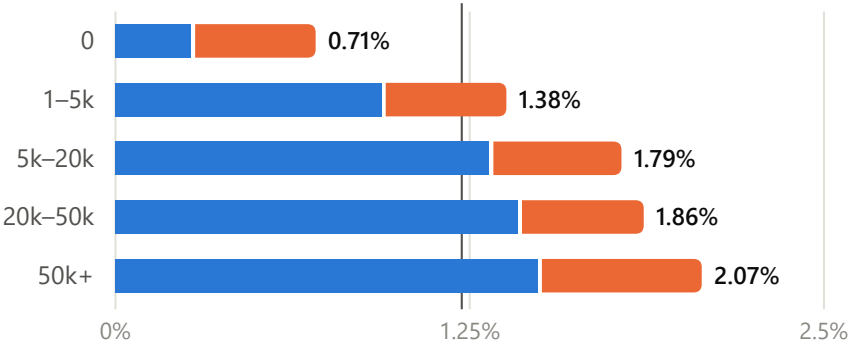
So what: behaviour beats demographics. Use account activity as the first targeting filter.

Active accounts convert 3× better than dormant ones

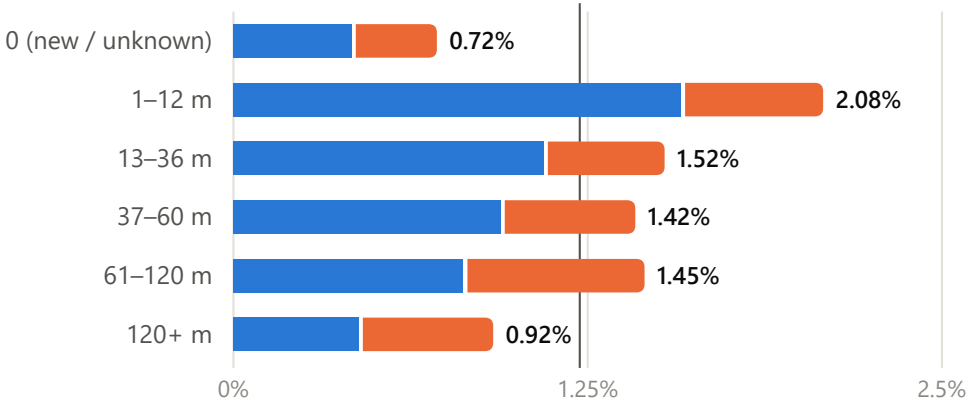
Acceptance rate split by product · vertical line = 1.22% average

PA insurance Life insurance

Cash inflow, last 30 days (THB)



Months on book (tenure)



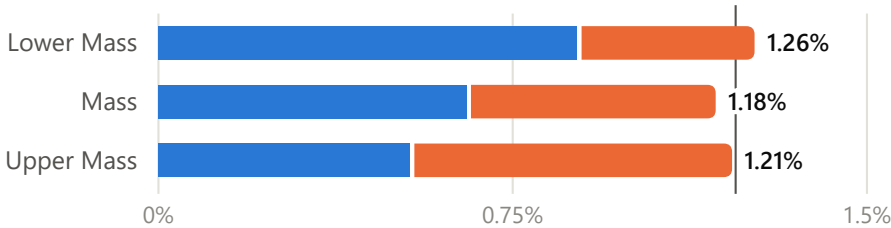
So what: half of the customers had **no inflow** and accept at 0.71%, against **2.07%** at 50k+. The gap is almost all PA. New accounts (1-12 months) are the best tenure group at 2.08%.

Match the product: Life for the affluent, PA for the mass market

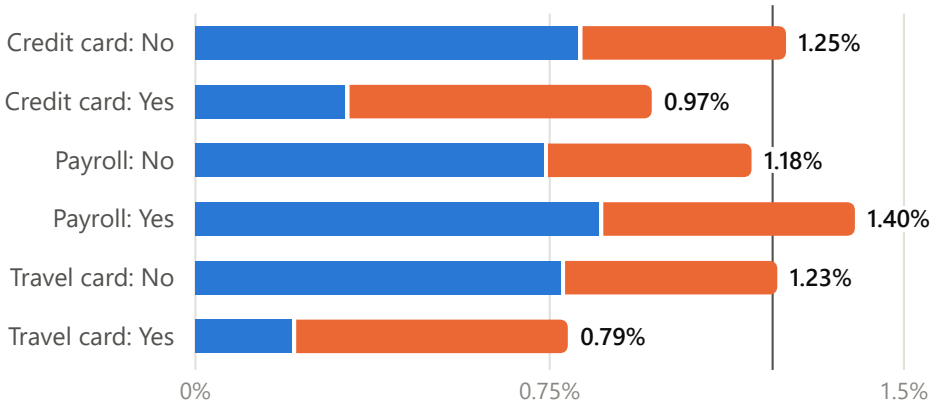
Same total acceptance in every segment (~1.2%), but a very different product mix

PA insurance Life insurance

Customer segment



Product holding



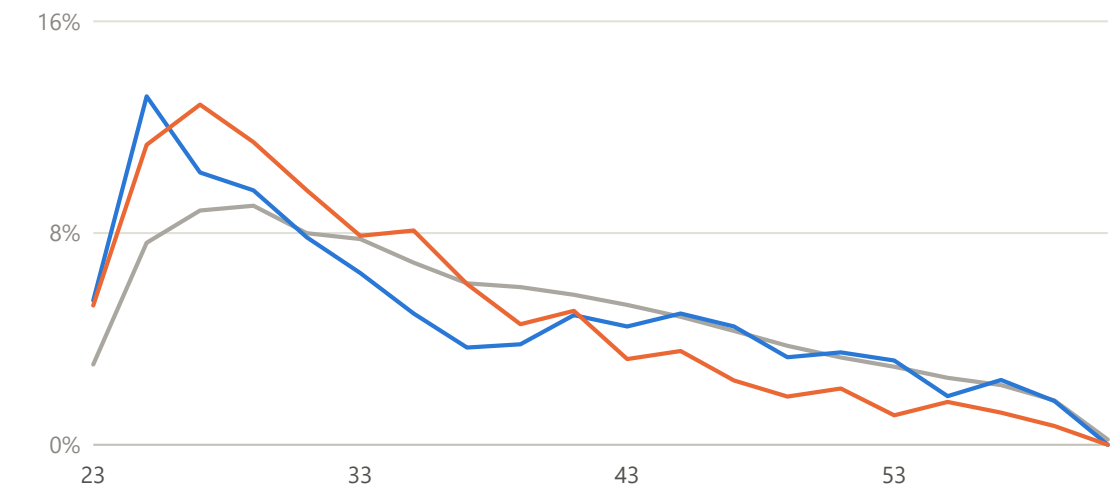
So what: Upper Mass buys Life more than PA (0.68% vs 0.53%), and so do credit-card holders (0.65% vs 0.32%). Lower Mass is mostly PA. Offer each customer the right product.

Buyers are younger; Life buyers earn more and have longer tenure

How buyers differ from customers who rejected

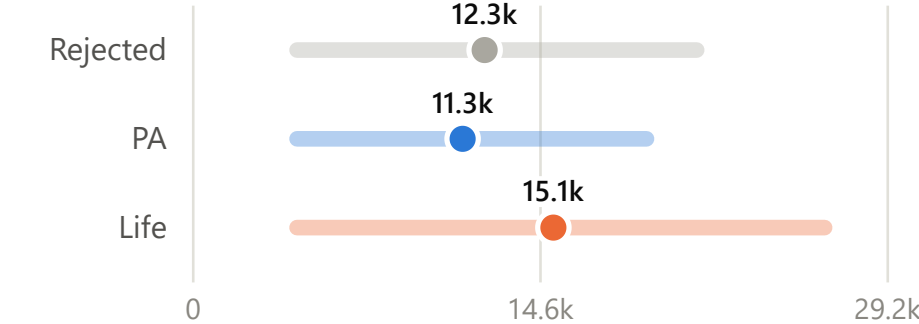
Age distribution by outcome

Rejected PA buyers Life buyers

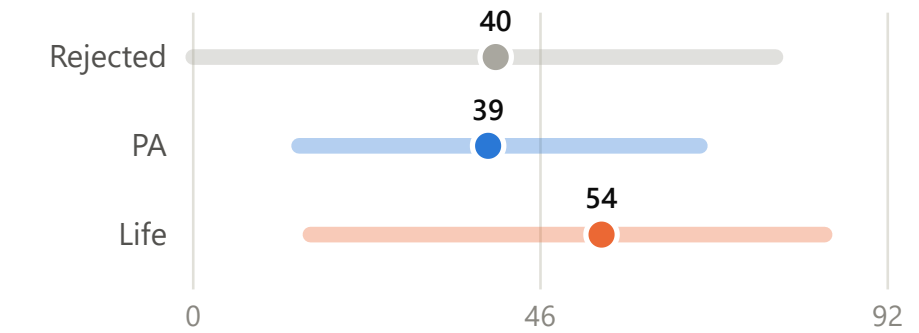


Median (dot) and middle 50% (band)

Monthly income (THB)



Months on book



So what: PA buyers peak at age 25 and Life buyers at 27, against 29 for rejecters. Life buyers' median income is 15.1k (vs 12.3k) and median tenure is 54 months (vs 40).

A model reaches half of the buyers with a fifth of the calls

Gradient-boosted trees · trained on 150k customers, tested on 64k it had never seen

ROC-AUC

0.72

0.5 = random

Top 20% reach

49%

of buyers

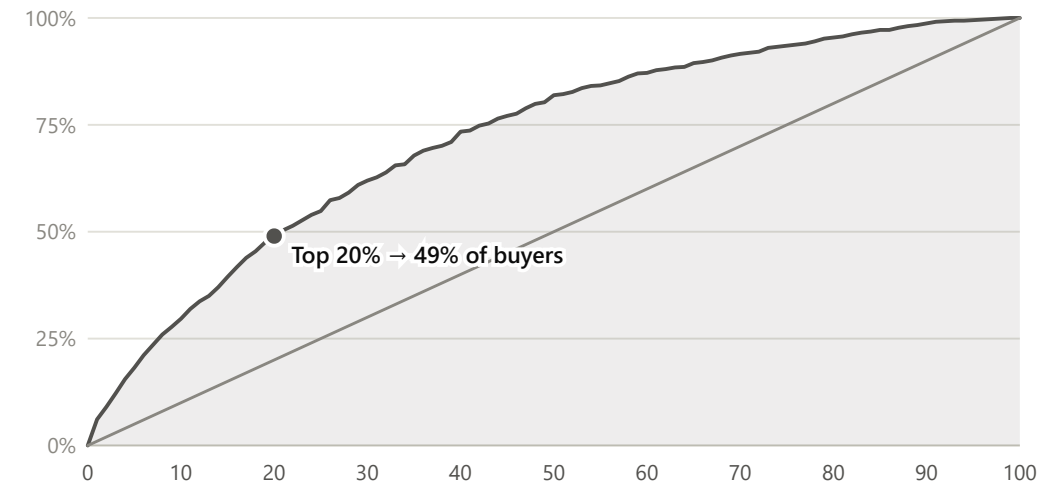
Top 10% lift

3.0×

3.6% vs 1.2%

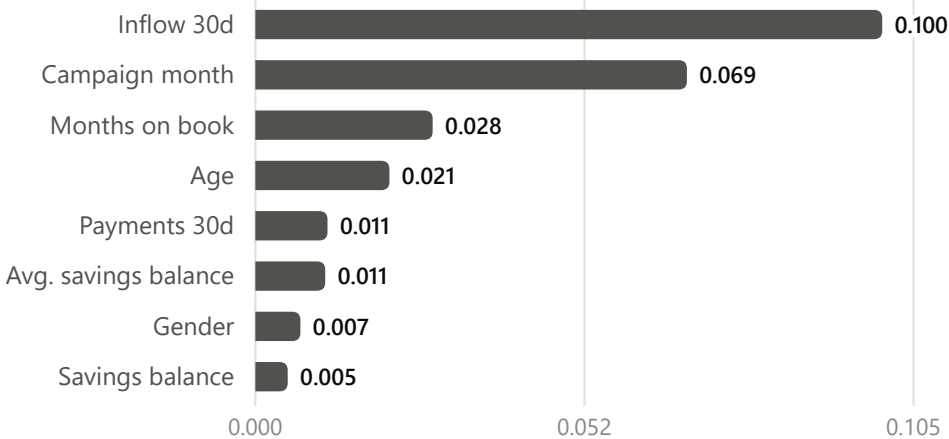
Cumulative gains

— Model — Random targeting



What the model relies on

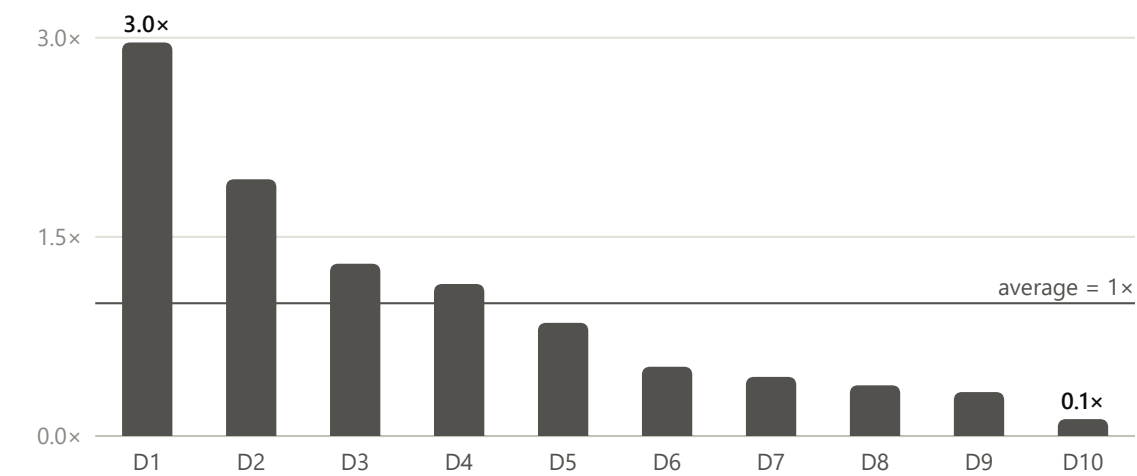
Drop in ROC-AUC when the feature is shuffled



From score to action: three contact tiers

Rank customers by score, cut into 10 equal slices (deciles), then group them into treatments

Lift by score decile (D1 = highest scores)



Suggested contact tiers

Tier	Customers	Acceptance	Lift	Share of buyers	Treatment
A · Call first	Top 0–20%	2.99%	2.4×	49%	Agent call, best offer
B · Standard	Top 20–50%	1.34%	1.1×	33%	Standard campaign
C · Low priority	Top 50–100%	0.44%	0.4×	18%	Digital only or skip

Tier A (top 20%) converts at 3.0% and holds half of the buyers. **Tier C** (bottom 50%) holds only 18%, so move it to SMS or app push, or skip it.

Five actions for the next campaign

Ordered by expected impact

1

Score before every campaign

Rank customers with the propensity model and call Tier A first.

Top 30% → 62% of buyers

2

Match the product

Life to Upper Mass, income $\geq 30k$, credit-card holders; PA to the active mass market.

Higher conversion per call

3

Prioritise active & young

Inflow $> 5k$, 1–36 months on book, age ≤ 30 , savings 1k–20k.

1.5–2.1% vs 1.2% average

4

Fewer, better contacts

Cut low-score volume in the big months instead of mass contact.

Jun–Sep: $\sim 2\times$ the rate

5

Fix the data

Repair the 15-day flows and tenure; add a customer ID plus the Policy and Payment data.

Better model, persistency view

Start with 1 and 2: they need no new data and can run in the next campaign as an A/B test, with model-targeted tiers plus product matching against current targeting, so the gain is measured, not assumed.

What to keep in mind, and what comes next

Honest limits of this analysis

Limitations

- No customer ID: 2,173 duplicate rows cannot be confirmed as real repeats.
- No year on campaign month, so seasonality and trend cannot be separated.
- 26% of customers have tenure = 0 (likely unknown); 839 have an impossible tenure.
- Policy and Payment data were not provided, so there is no view after the sale.
- Correlation, not causation: the model ranks customers, it does not prove why they buy.

Next steps

- **A/B test** model-targeted vs current targeting in the next campaign.
- Add a **product-choice model** (PA vs Life) on top of the buy / no-buy score.
- Validate on a **later time period**, not only a random hold-out.
- Join **Policy and Payment** data to optimise for persistency, not just the first sale.
- Put the pipeline on a **monthly schedule** with data-quality alerts.

THANK YOU

Questions?

Interactive dashboard: **Answer Case 1A.html** (hover any chart, filter by segment) · code:
src/campaign_analysis · run: **python main.py**